

PRATYASHA BHAKUNI

OBJECTIVE

To utilize my education, skills and knowledge for a purposeful result which outstand me for a challenging position and adds growth to the organization's motto. I am a self-motivated achiever and a hard-working person with an ability to plan and execute things efficiently and dedicatedly believing in smart work.

PERSONAL DETAILS

✉ pratyashabhakuni29@gmail.com

☎ +91 8937999348

📍 Noida, Uttar Pradesh, India

🌐 <https://www.linkedin.com/in/pratyasha-bhakuni-1b669a168>

GENDER : Female

NATIONALITY : Indian

DATE OF BIRTH : 29 December'1999

LANGUAGE PROFICIENCY : English and Hindi

PROJECTS

1. MORPHOMETRIC ANALYSIS AND DISTRIBUTION OF FRESHWATER MOLLUSCS LAMELLIDENS MARGINALIS

This project represents the growth parameters and viability of freshwater mussel and helps to understand the favorable conditions for the survival of *Lamellidens marginalis* and its growth rate under observance at timely gap of certain days. This has also helped in understanding the vulnerability of freshwater mussel in the present ecological imbalance and environmental conditions.

UNDER : DR. VIVEKANAND BAHUGUNA, ASSISTANT PROFESSOR, UTTARANCHAL UNIVERSITY,
DEHRADUN

DATED : 1 FEB 2021 – 5 JUN 2021

2. SEQUENCE ANALYSIS AND ALIGNMENT OF GENES RESPONSIBLE FOR PHENYLKETONURIA DISEASE IN HOMO SAPIENS

This project involves understanding the metabolic pathway of **Phenylketonuria** and the genes responsible for the disease. It also covers its sequence analysis and construction of phylogenetic tree using basic bioinformatics tools.

UNDER : DR. KUMUND SARIN, BIOINFORMATICS INSTITUTE OF INDIA, NOIDA

DATED : 1 JAN 2021 – 29 JAN 2021

3. DEVELOPMENT AND VALIDATION OF IN-VITRO POTENCY OF RECOMBINANT FOLLICULAR STIMULATING HORMONE

This project involves developing and validating the potency of different batches of recombinant Follicular Stimulating Hormone obtained from different manufacturers in in-vitro conditions using **SANDWICH ELISA** along with its protein estimation first by **LOWRY METHOD** and then using **HIGH PERFORMANCE LIQUID CHROMATOGRAPHY (HPLC)**

UNDER : DR. CHARU M. KAMAL, SCIENTIST GRADE- I, ENZYMES & HORMONES LABORATORY,
NATIONAL INSTITUTE OF BIOLOGICALS, NOIDA
DATED : 1 JAN 2023 – MAY 2023

SKILLS & ABILITIES

-  **MOLECULAR BIOLOGY TECHNIQUES** : Plasmid DNA isolation and DNA quantitation, Restriction digestion, Preparation of competent cells, Agarose gel Electrophoresis, DNA Ligation, Polymerase Chain Reaction, Restriction mapping of recombinant plasmid, RFLP analysis
-  **ANALYTICAL TECHNIQUES** : Spectrophotometry, Gel Documentation, ELISA, Centrifugation, SDS-PAGE of BSA and extracted proteins, Paper chromatography of amino acids, Extraction and Purification of plant and animal protein, T.L.C. of lipids, Gel Permeation and ion exchange chromatography
-  **MICROBIOLOGY AND BIOPROCESSING TECHNIQUES** : Sterilization, disinfection of laboratory for accurate results, Preparation of media for growth of various microorganisms, Gram staining and enumeration of microorganisms, Storage of microorganisms, Isolation and maintenance of organisms by Plating, Streaking and Serial Dilution methods, Maintenance of bacterial population by turbidometry, Isolation and screening of industrially important microorganisms
-  **GENETIC ENGINEERING** : PCR amplification of genomic region of cell, Isolation of genomic DNA, Induction of recombinant protein with IPTG
-  **BIOCHEMICAL ANALYSIS** : Titration of amino acids, Quantitative estimation of proteins and sugars
-  **BIOINFORMATICS** : Access to biological databases and sequences, structure and strain analysis using genetic tree, sequence analysis of DNA, RNA and proteins, BLAST and FASTA for local alignment of nucleotide and amino acid sequences
-  **SOFTWARE SKILLS** : Proficient with Word processing,

Spreadsheets and Presentation software, MS Outlook and MS Excel

CONFERENCES, WORKSHOPS & OTHER PARTICIPATION

- ☐ Training workshop for capacity building of educational institutions about water quality with special reference to water safety and security
- ☐ National Science Day program by UCOST
- ☐ Disaster management event conducted by NDRF
- Health awareness and support program by Hans Foundation
- Intellectual Property training program under National Intellectual Property Awareness Mission
- Speech competition on Importance of Right to Vote in a Democracy
- Cancer awareness seminar on World Cancer Day
- Training program on Quality Control and Analytical Techniques
- Workshop on Research Practices and Design for professionals
- War of Words event organized by NIRMAAN society

EDUCATION

01/07/2023

**M.Sc. BIOTECHNOLOGY |SIR J.C. BOSE COLLEGE OF BIOTECHNOLOGY,
KUMAUN UNIVERSITY|**

SEMESTERS	SUBJECTS STUDIED	GRADE POINT
FIRST	Biochemistry, Molecular Biology, Cell & Developmental Biology, Microbiology And Industrial Applications, Biostatistics And Computer Applications	8.68
SECOND	Plant Biotechnology, Analytical Techniques, Genetic Engineering, Molecular Genetics, Genomics And Proteomics	8.09
THIRD	Bioprocess Engineering & Technology, Environmental Biotechnology, Animal Biotechnology, Immunology and Immunotechnology, Molecular Virology	Not declared

12/07/2021

B.Sc. (Hons.) BIOTECHNOLOGY | SCHOOL OF APPLIED & LIFE SCIENCES,
UTTARANCHAL UNIVERSITY |

SEMESTERS	SUBJECTS STUDIED	GRADE POINT
FIRST	Biochemistry & Metabolism, Cell Biology, Biotechnology And Human Welfare, Remedial Mathematics, E.V.S.	8.40
SECOND	Mammalian Physiology, Microbiology And Plant Physiology, Developmental Biology, English Communication, Biostatistics & Computational Biology	8.73
THIRD	Genetics, General Microbiology & Metabolism, Chemistry- I, Bioethics And Biosafety, Enzymology	8.23
FOURTH	Molecular Biology, Immunology, Chemistry-II, Entrepreneurship Development, Molecular Diagnostics	9.42
FIFTH	Industrial Fermentation, R.D.T. , Bioinformatics, Environmental Biotechnology	8.04
SIXTH	Bioanalytical Tools & Techniques, Genomics And Proteomics, Plant Biotechnology, Animal Biotechnology	9.00

CLASS XI |2017| NIRMALA CONVENT SENIOR SECONDARY SCHOOL, KATHGODAM| CBSE|
70.7%

CLASS X |2015 |ST. FRANCIS CONVENT SCHOOL, BAREILLY| CBSE | 85.5%

RESEARCH INTERESTS

- **BIOFUEL AND BIO-ENERGY GENERATION USING GENETIC MODIFICATION IN PLANTS**
- **TRANSGENIC PLANTS FOR XEROPHYTIC CLIMATIC CONDITIONS**
- **STEM CELL TECHNOLOGY FOR ENDANGERED SPECIES**
- **ENVIRONMENTAL BIOTECHNOLOGY IN ENSURING NET ZERO EMISSIONS FOR FUTURE**
- **NANOTECHNOLOGY FOR MEMORY TRANSITION AND MEMORY CHIPS**
- **NEUROSCIENCE**

ACTIVITIES

1. **Delivered presentations during seminars on the topics**

- ☐ Protein crystallization of complex proteins
- ☐ Co- infection of Influenza A virus and Respiratory syncytial virus(RSV)
 2. Hosted the college SCIENTIFIC ANNUAL MEET and cultural program (2022)
 3. Dengue Awareness Campaign
 4. Cancer Awareness Program
 5. HIV Awareness Program
 6. Participated in INDIAN LEADERSHIP SUMMIT 2.0

LEADERSHIP

I was the leader of Dengue Awareness campaign held in Dehradun in the year 2021

ADDITIONAL INFORMATION

- 1 I am currently pursuing hands-on-training in Quality Control Standards, Enzymes & Hormones Laboratory at National Institute of Biologicals
- 2 I am currently pursuing Diploma in women's empowerment and development from IGNOU
- 3 I am a trained Kathak classical dancer
- 4 I have keen interest in exploring applications of biotechnology in military field

CO- CURRICULAR & HOBBIES

- ☐ Reading research papers on biotechnology in Defence
- ☐ Reading fictional books
- ☐ Writing articles and blogs
- ☐ Poster making & painting

REFERENCES

DR. VEENA PANDE, PROFESSOR – HEAD OF DEPARTMENT, DEPARTMENT OF BIOTECHNOLOGY, SIR J.C. BOSE COLLEGE, KUMAUN UNIVERSITY

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DR. TAPAN K. NAILWAL, PROFESSOR, DEPARTMENT OF BIOTECHNOLOGY, SIR J.C. BOSE COLLEGE, KUMAUN UNIVERSITY

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9412986483

**DR. INDRA RAUTELA, ASSISTANT PROFESSOR (G-II) - DEPARTMENT OF
BIOTECHNOLOGY, UTTARANCHAL UNIVERSITY**

indrarautela@gmail.com

8430633253

DECLARATION I hereby declare that all the above information given by me is true to the best of my knowledge and belief.

Date : 30.04.2023

Place : Noida